

The Costs of Production

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A companion to the lecture slides. Each slide that hides a step is reproduced here and worked through line by line.

Before we begin

Required readings

Required readings

The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 14** — The Costs of Production.

Read the chapter alongside these slides.

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Slide 2

Working through it

Why this chapter. Mankiw's chapter 14 covers the same ground in the same order: what economists count as a cost, the production function and diminishing marginal product, the short-run cost measures, and long-run average cost with economies and diseconomies of scale.

What the slides add. The main line here is *calculus-free*. Marginal cost is built as a difference in the total-cost column, marginal product as a difference in the output column, and the

result that marginal cost crosses average cost at the minimum is demonstrated *in a table* using the pulling principle rather than asserted or differentiated. A single marked extension frame gives the calculus version afterwards, for readers who want it.

How to use it. Read the chapter once before and once after. The chapter will tell you the curves have these shapes; the slides show you where the shapes come from, which is what makes them impossible to forget.

Where we are going

Where we are going

1. **What a firm's costs really are** — the economist counts what the accountant leaves out.
2. **The production function** — workers in, components out, and each worker adds less.
3. **From production to cost** — why the cost of one more unit keeps climbing.
4. **The family of cost measures** — total, fixed, variable, average, marginal.
5. **Marginal pulls average** — the crossing that every exam asks about.
6. **The long run** — when the factory itself is up for grabs.
7. **Pulling it together** — one chain from technology to cost curves.

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Slide 3

Working through it

The shape of the argument. Seven parts, and they form one chain rather than seven topics.

Part 1 settles what counts as a cost. The economist's answer is larger than the accountant's, and the difference changes decisions.

Parts 2–3 build a small factory. A production function turns workers into output; because one input is fixed, each worker adds less than the last; and that single fact, priced at a wage, makes the cost of each extra unit climb.

Parts 4–5 slice one cost curve into the standard family of measures and then establish the central result: marginal cost pulls every average toward itself, and therefore crosses each at its minimum.

Parts 6–7 free the fixed input, so the firm chooses its factory, and then assemble the whole chain.

The thread. Everything is derived from one production function and one cost function. Nothing is assumed about the shape of a curve.

1. Part 1: What a Firm's Costs Really Are

Why open the firm up?

Why open the firm up?

The supply curve summarized firm behavior: a higher price, a larger quantity supplied. It never said **why**.

The why lives in the firm's costs. They decide three things:

- whether the **next unit** is worth producing,
- what **price** the firm must fetch to survive,
- how the firm behaves as it **grows**.

Behind every supply decision is a cost schedule.

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Slide 5

Working through it

What this frame is for. It names the gap that the rest of the material fills. The supply curve has been used as a summary of firm behavior without ever being explained.

What the supply curve said, and did not. It said quantity supplied rises with price. It never said why, how steeply, or where the curve comes from.

Where the answer lives. In the firm's costs. A manager deciding whether to run a second shift does not consult a curve; they compare what the extra output will fetch against what it will cost. So the curve is a *consequence* of a cost schedule, and the cost schedule is the primitive object.

The three decisions costs govern.

- Whether the **next unit** is worth producing — which is the supply decision itself.
- What **price** the firm must fetch to survive — which determines entry and exit.
- How the firm behaves as it **grows** — which determines how many firms an industry can hold.

Each is taken up in turn.

The goal: profit

The goal: profit

Profit. Total revenue minus total cost: $\pi = TR - TC$, where $TR = P \times Q$.

Our maintained assumption: **firms maximize profit.**

Not because owners think of nothing else — but because profit maximization predicts observed firm behavior remarkably well.

Revenue is the easy half. The work ahead is the other half: **what belongs in total cost?**

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Slide 6

Working through it

The objective. $\pi = TR - TC$, where $TR = P \times Q$.

The assumption, and why it is defensible. Firms are taken to maximize profit. This is not a claim that owners think about nothing else. It is a claim that the *predictions* generated by assuming it match what firms actually do — which is the standard by which any modeling assumption should be judged.

Why revenue is the easy half. $P \times Q$ is observable and uncontroversial. Both terms appear on invoices.

Why cost is the hard half. Reasonable people disagree about what belongs in it. An accountant preparing a tax return counts one set of items; an economist assessing the same

firm counts a larger set. The disagreement is not a technicality: the two answers can differ by a factor of four, as the frames ahead demonstrate, and they can point to opposite decisions about whether to stay in business.

So the first job. Settle what goes into TC before computing anything with it.

Two kinds of cost

Two kinds of cost

Explicit costs. Input costs that require a direct money payment: wages, rent, materials, interest on a loan.

Implicit costs. Input costs with no money payment — the value of what the owner gives up: the salary not earned elsewhere, the interest the invested savings could have earned.

An **opportunity cost** is whatever is given up to get something. **The economist counts both kinds. The accountant counts only the explicit.**

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Working through it

The organizing rule. The cost of anything is what you give up to get it — its **opportunity cost**. Everything in this part follows from taking that seriously.

Explicit costs. Input costs requiring a direct money payment: wages, rent, materials, interest on a loan. Nobody disputes these. They leave a paper trail, and the accountant records them.

Implicit costs. Input costs with no money payment — the value of what the owner gives up. The salary not earned elsewhere. The interest the invested savings would have earned. The rent forgone on a building the owner already owns.

Why the accountant excludes them, and is not being careless. Accounting rules require verifiable transactions. An implicit cost has no invoice, no counterparty and no audit trail, so it cannot be recorded under those rules — sensibly, for tax and reporting purposes.

Why the economist includes them. Because the question is different. An accountant asks what happened to the money. An economist asks whether this use of resources beats the alternatives — and answering that requires valuing the alternatives, which is exactly what implicit costs are.

The engineer who left

The engineer who left		
A software engineer leaves a \$150,000 job to consult on her own: office rent, equipment, and \$200,000 of savings that had been earning 5 percent.		
	Item	\$/year
Explicit	office rent	30,000
Explicit	equipment	20,000
Implicit	salary given up	150,000
Implicit	interest given up: $0.05 \times 200,000$	10,000
Total opportunity cost		210,000
The accountant sees \$50,000. The economist sees \$210,000 .		
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Working through it

The situation. A software engineer leaves a \$150,000 job to consult independently. She rents an office, buys equipment, and puts \$200,000 of her own savings into the business. Those savings had been earning 5 percent.

The explicit costs. Office rent \$30,000; equipment \$20,000. Total **\$50,000** — and this is where the accountant's ledger stops.

The implicit costs.

- **Salary given up:** \$150,000. She is working full-time in her own firm, and every hour spent there is an hour not sold to an employer.
- **Interest given up:** $0.05 \times \$200,000 = \$10,000$. The savings did not vanish, but they stopped earning.

The total.

$$50,000 + 150,000 + 10,000 = \textbf{\$210,000 per year}$$

The size of the gap. The accountant sees \$50,000; the economist sees \$210,000 — more than four times as much. Note that no fact about the business is in dispute. Both numbers are correct answers to different questions, and only the second one can tell her whether consulting beats her old job.

The cost of capital: own it or borrow it

The cost of capital: own it or borrow it

A firm buys \$500,000 of equipment. The interest rate is 5 percent.

- **Borrowed:** the firm pays \$25,000 interest per year — an **explicit** cost.
- **Owner's own money:** no payment leaves — but the money stops earning \$25,000 elsewhere — an **implicit** cost.

Same equipment, same \$25,000 economic cost either way.

Only the **accounting** treatment differs.

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Working through it

Why this frame exists. It is the cleanest possible test of the implicit cost idea, because it holds *everything* constant except the financing.

The setup. A firm acquires \$500,000 of equipment. The interest rate is 5 percent.

Financed by borrowing. The bank charges $0.05 \times 500,000 = \$25,000$ a year. This is explicit: a payment leaves, an invoice exists, the accountant records it, and it is tax-deductible.

Financed from the owner's own funds. No payment leaves. But \$500,000 that was earning 5 percent somewhere else has stopped earning it — a sacrifice of exactly \$25,000 a year. This is implicit: no invoice, no deduction, and invisible in the accounts.

The point. **Same equipment, same \$25,000 of economic cost, either way.** The physical resources committed are identical and the sacrifice is identical. Only the *accounting treatment* differs.

Why that matters practically. A firm comparing itself to a rival that borrows will look more profitable on paper purely because of how it financed the same machines. Any comparison of the two that uses accounting profit is comparing financing decisions, not performance.

Two profits

Two profits

Accounting profit. Total revenue minus explicit costs.

Economic profit. Total revenue minus **all** opportunity costs, explicit and implicit.

The consulting firm earns \$280,000 in revenue:

	Accounting	Economic
Revenue	280,000	280,000
Costs counted	50,000	210,000
Profit	230,000	70,000

The gap between them is **exactly the implicit costs**.

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Working through it

Two costs give two profits.

- **Accounting profit** = revenue – explicit costs.
- **Economic profit** = revenue – *all* opportunity costs.

The consulting firm's numbers. Revenue \$280,000.

$$\text{accounting: } 280,000 - 50,000 = \$230,000$$

$$\text{economic: } 280,000 - 210,000 = \$70,000$$

The gap is never mysterious. $230,000 - 70,000 = \$160,000$, which is precisely the implicit costs ($150,000 + 10,000$). By construction the difference between the two profits is always exactly the implicit costs — a useful check when both are computed.

Which is right? Both, for their own purposes. Accounting profit is what gets taxed and reported. Economic profit is what tells the owner whether to keep doing this rather than something else. Confusing them is not a rounding error: a firm can post a large accounting profit while paying its owner less than she could earn elsewhere, which is a business that should close.

What zero economic profit means

What zero economic profit means

A firm with **zero economic profit** covers every opportunity cost — including the owner's forgone salary and a normal return on the invested capital.

The owner is doing **exactly as well** as in the next-best alternative. Nothing is wrong.

- Economic profit > 0 : better than the alternative — others will want in.
- Economic profit < 0 : worse than the alternative — over time, the owner exits.

Economic profit, not accounting profit, steers decisions.

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Working through it

Why the phrase needs unpacking. “Zero profit” sounds like failure. In economic terms it is not.

What zero economic profit actually means. The firm has covered every opportunity cost — rent, wages, materials, *and* the owner's full market salary, *and* a normal return on all capital invested — and has nothing left over. The owner is doing exactly as well as in her next-best alternative. Nothing is wrong, and there is no reason to leave.

Reading the other two cases.

- **Economic profit** > 0 : this use of resources beats the alternatives. The owner stays, and others notice and want in.
- **Economic profit** < 0 : the alternatives are better. The firm may survive for a while — it can still be posting an accounting profit — but over time the owner exits.

Why this matters beyond definitions. Positive economic profit attracts entry and negative economic profit drives exit. That is the mechanism which pushes a competitive industry toward zero economic profit in the long run, and it is why the concept is worth getting exactly right here rather than later.

2. Part 2: The Production Function

Short run and long run

Short run and long run

Short run. At least one input is fixed. The firm can hire workers at short notice; it cannot build a bigger factory at short notice.

Long run. All inputs are variable — factory size included.

Parts 2–5 live in the **short run**: machines and floor space fixed, workers variable.

Part 6 opens the long run.

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Working through it

The distinction, and it is about time rather than clock hours.

- **Short run**: at least one input is fixed. The firm can hire workers quickly; it cannot pour a new factory floor quickly.
- **Long run**: all inputs are variable, factory size included.

Why the definition is about which inputs rather than about a number of months. A food truck's long run might be six weeks; a nuclear plant's might be a decade. The horizon that counts is however long it takes for the fixed input to become choosable.

Why this organizes everything. The short run is where diminishing marginal product operates, because diminishing marginal product requires something to be fixed for the variable input to be crowded against. No fixed input, no crowding, no diminishing returns — so the entire cost story of Parts 2 to 5 depends on being in the short run.

The plan. Parts 2 to 5 hold the factory fixed and vary labor. Part 6 lifts that and lets the firm choose the factory too, which changes the shape of the answer completely.

Write the equation

Write the equation

A firm assembles electronic components. Machines and floor space are fixed; it chooses only L , the number of workers per shift.

Production function. The relationship between inputs used and output produced: here $Q = f(L)$.

$$Q = 24L - L^2 \quad \text{for } 0 \leq L \leq 12$$

24 is the pace a worker sets with the machines to themselves; $-L^2$ is the crowding penalty that grows as the floor fills. Past $L = 12$, another worker would **lower** output.

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Slide 16

Working through it

The firm. An electronic-components assembler. Machines and floor space are fixed; the only choice is L , workers per shift.

The definition. A **production function** relates inputs used to output produced: $Q = f(L)$.

The function.

$$Q = 24L - L^2, \quad 0 \leq L \leq 12$$

Reading each term, which is the habit worth building.

- $24L$: the pace one worker sets with the machines to themselves. Twenty-four units per worker, if there were no interference.
- $-L^2$: the crowding penalty. It is squared, so it grows faster than the workforce — doubling the workers quadruples the interference. This single term is what produces every result in the next three parts.

Why the range stops at 12. Beyond $L = 12$ the crowding term dominates and an extra worker would *lower* total output. No sensible manager hires there, so the function is stated only where it is economically meaningful.

Plug the numbers in

Plug the numbers in		
Workers L	$24L - L^2$	Output Q
0	$24(0) - 0$	0
1	$24(1) - 1$	23
2	$24(2) - 4$	44
3	$24(3) - 9$	63
4	$24(4) - 16$	80
5	$24(5) - 25$	95
6	$24(6) - 36$	108
8	$24(8) - 64$	128
12	$24(12) - 144$	144

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Slide 17

Working through it

Every row substituted, so nothing rests on trust.

$$L = 1 : 24 - 1 = 23 \quad L = 5 : 120 - 25 = 95 \quad L = 12 : 288 - 144 = 144$$

What to read off the output column. Output rises throughout — every worker adds something. But the *increments* shrink visibly: the first worker brings 23 units, while going from

8 workers to 12 (four more people) brings only $144 - 128 = 16$.

The distinction that matters, stated now because it is the most common confusion in the whole topic. Output rising and additions shrinking are compatible. A firm can be producing more with every hire while each hire contributes less than the previous one. Diminishing marginal product is a claim about the second thing, never the first.

Where the table is going. The next frame turns the shrinking increments into a column of their own, and the frame after that names the pattern.

Marginal product: what the next worker adds

Marginal product: what the next worker adds

Marginal product. The extra output from one more unit of input, other inputs held fixed: $MP = \Delta Q / \Delta L$.

The manager's question: **if I add one more worker, how many more components do I get?**

It is a subtraction on the output table. The first worker: $23 - 0 = 23$. The second: $44 - 23 = 21$.

Now run it all the way down...

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Slide 18

Working through it

The definition. **Marginal product** is the extra output from one more unit of input, other inputs held fixed: $MP = \Delta Q / \Delta L$.

Why it is the manager's question. Nobody hiring asks what workers produce on average. They ask: if I add one more worker, how much more output do I get, and is it worth the wage? Marginal product answers the first half.

No new formula is required. It is a subtraction on the output table already built:

$$\text{1st worker: } 23 - 0 = 23 \quad \text{2nd worker: } 44 - 23 = 21$$

The phrase “other inputs held fixed” is doing real work. Marginal product is defined *at a given factory*. Add a second machine at the same time as the worker and the number would be different — which is precisely why the long run in Part 6 changes the picture.

Already visible after two rows. 23, then 21. The additions have started shrinking, and the next frame runs the subtraction all the way down.

The marginal product column

The marginal product column		
L	Q	MP of that worker
1	23	$23 - 0 = 23$
2	44	$44 - 23 = 21$
3	63	$63 - 44 = 19$
4	80	$80 - 63 = 17$
5	95	$95 - 80 = 15$
6	108	$108 - 95 = 13$
7	119	$119 - 108 = 11$
8	128	$128 - 119 = 9$

...and so on down: the 12th worker adds just 1.

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Slide 19

Working through it

The column, built by subtraction. 23, 21, 19, 17, 15, 13, 11, 9 — and continuing down to 1 for the twelfth worker.

The pattern. Every worker adds something positive, and every worker adds *exactly two less* than the one before.

Where the constant gap of 2 comes from. It is the $-L^2$ term. The difference between $24L - L^2$ at consecutive values of L falls by 2 each time, because squaring is what makes the penalty accelerate. The linear production term contributes a constant 24; the squared term subtracts a steadily larger amount.

The important observation. Nobody chose this pattern or imposed it. It was not an assumption about worker quality or motivation. It fell out of one algebraic term in a function

written down two frames ago — which is what makes the result in the next frame a derivation rather than a claim.

Sanity check available to you. The marginal products should sum to total output. $23 + 21 + 19 + \dots$ down to the twelfth worker's 1 gives 144, the output at $L = 12$. Active Learning 2 uses exactly this check.

Diminishing marginal product

Diminishing marginal product

Diminishing marginal product. As more of a variable input is added, holding other inputs fixed, its marginal product eventually falls.

Why: the **fixed input gets stretched**. Early workers have machines to themselves; later workers share, wait, and coordinate around the same floor.

Every worker adds output — and each adds less than the one before.

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Slide 20

Working through it

The definition. As more of a variable input is added, holding other inputs fixed, its marginal product *eventually* falls.

The cause, and it is always the same one. The fixed input gets stretched. The first worker has every machine available; the ninth waits for the soldering station. The workers are interchangeable; what changes is how much fixed capital each of them has to work with.

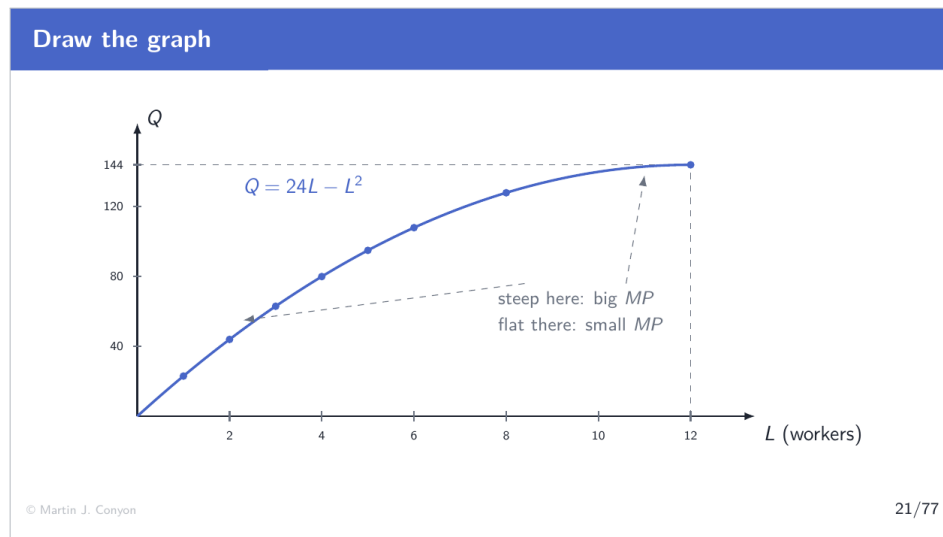
Two things the law does not say, and both are commonly misread.

- It does **not** say later workers are less capable. Swap the first and ninth workers and nothing changes — the ninth position is the problem, not the person.
- It does **not** say output falls. Output keeps rising throughout the relevant range. It is

the *additions* that shrink.

Why the word “eventually” is in the definition. Some production processes show *increasing* marginal product at first — two workers can do things one cannot, such as holding and cutting simultaneously. The law claims only that this cannot continue: with something fixed, diminishing returns arrive sooner or later.

Draw the graph



Slide 21

Working through it

What is plotted. Workers horizontally, output vertically. The dots are the table’s rows and the curve is the function — and they coincide, as they must, because both came from $Q = 24L - L^2$.

Reading the shape, which is the whole purpose of the frame. The *slope* of this curve at any point is marginal product: it is extra output per extra worker.

- **On the left the curve is steep.** Each additional worker lifts output sharply — the large marginal products of 23, 21, 19.
- **On the right it is nearly flat.** The eleventh and twelfth workers barely lift it — marginal products of 3 and 1.

So *diminishing marginal product is a visual property*. It is exactly the *flattening* of this curve. Not an extra assumption laid on top of the picture, but a description of the picture's shape.

Hold the image. In Part 3 the same fact appears as a cost curve that *steepens*. The two shapes are mirror images of one another, and seeing that connection is the point of the "Mirror image" frame.

The margin is where the decision lives

The margin is where the decision lives

The manager never asks "how productive are workers on average?"

The manager asks: **what does one more worker add, and what does one more worker cost?**

- One more worker **adds** *MP* components.
- One more worker **costs** the wage w .

Next: put a dollar figure on the fixed bill and the wage, and the cost curves build themselves.

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Slide 22

Working through it

How a manager actually reasons. Not about averages. Nobody staffing a shift asks what the average worker has historically produced, because that number is about workers already hired and the decision is about the *next* one.

The two halves of the hiring decision.

- One more worker **adds** *MP* units of output.
- One more worker **costs** the wage w .

What Part 2 has supplied. The first half, completely: the next worker adds the marginal product, and that number shrinks as the floor fills.

What is still missing. A price. Marginal product is measured in components, and the wage in dollars, so the two cannot yet be compared. Part 3 supplies the wage and the fixed bill, and the comparison becomes possible.

Why this framing matters beyond hiring. Marginal thinking — comparing the addition against the cost of the addition, one unit at a time — is the same move that produced the efficient quantity in earlier work and will produce the profit-maximizing output later. The object changes; the method does not.

3. Part 3: From Production to Cost

The firm's two bills

The firm's two bills

Back to the component assembler. Each shift it pays:

- **\$200** for machines, floor space, insurance — **owed even at zero output**;
- **\$50 per worker** hired.

$$TC = 200 + 50L$$

One bill is stuck; the other moves with the workforce. Costs inherit their shape **from the production function**.

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Slide 26

Working through it

The two bills.

- **\$200 per shift** for machines, floor space and insurance — owed even at zero output.
- **\$50 per worker** hired.

Total cost as a function of workers.

$$TC = 200 + 50L$$

Why that looks too simple. As a function of *workers* it is perfectly linear: every worker costs the same \$50, so cost rises in equal steps. There is nothing interesting in it.

Where the interest is. The firm does not sell workers; it sells output. So the object that matters is cost as a function of *output*, and that is where the production function re-enters. Because each successive worker produces less, equal steps in cost buy unequal steps in output — and cost as a function of output is therefore not linear at all.

The sentence to carry into the next frames. Cost curves inherit their shape from the production function. Nothing about the wage or the rent creates the curvature; the crowding term does.

Plug the numbers in

Plug the numbers in				
L	Q	Fixed (\$)	Labor $50L$ (\$)	TC (\$)
0	0	200	0	200
1	23	200	50	250
2	44	200	100	300
3	63	200	150	350
4	80	200	200	400
5	95	200	250	450
6	108	200	300	500
8	128	200	400	600
12	144	200	600	800

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Slide 27

Working through it

How the table is built. It is the production table with two money columns attached. Fixed cost is \$200 in every row. Labor cost is $50L$. Total cost is their sum.

What is constant and what is not. Total cost climbs in perfectly even \$50 steps — one step per worker. That column contains no curvature at all.

Now look at what the money buys.

- The first \$50 (one worker) buys **23** components.
- Going from 8 workers to 12 costs **\$200** and buys $144 - 128 = \mathbf{16}$ components.

The comparison. Four times the money, two-thirds of the output. Same wage per worker throughout; the difference is entirely on the output side.

What has just been demonstrated. Diminishing marginal product converted into dollars, row by row. The cost column is linear and the output column is not, so cost-per-unit-of-output must be rising — which is the subject of the rest of the part.

Rearrange to plot

Rearrange to plot

We want cost **against output** — TC as a function of Q , because output is what the firm sells.

Every row of the table is already a point (Q, TC) :

producing 63 components costs \$350 → $(63, 350)$
 producing 128 components costs \$600 → $(128, 600)$

For the smooth curve, the computer finds, for each Q , the workforce that produces it — then charges $200 + 50L$.

Working through it

The pivot. So far cost has been expressed as a function of workers. But workers are not what the firm sells, so the useful object is TC as a function of Q .

No new arithmetic is needed. Every row of the existing table already contains both numbers,

so every row is already a point (Q, TC) :

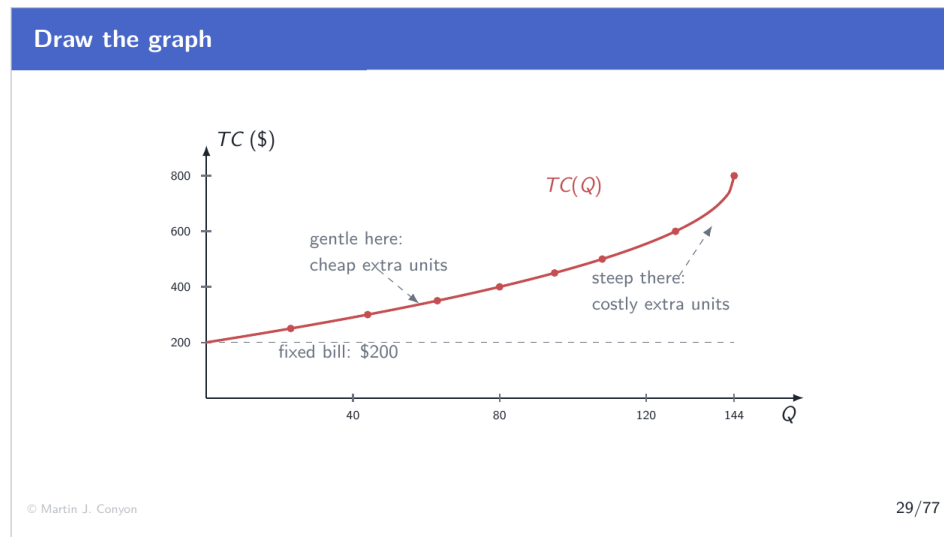
$$(63, 350) \quad (128, 600)$$

Producing 63 components costs \$350; producing 128 costs \$600.

What the smooth curve does between the dots. For each output level, it finds the workforce that just produces it — inverting $Q = 24L - L^2$ — and then charges $200 + 50L$. That is the same procedure as the table, applied at every output rather than at nine of them.

Why this is worth an explicit frame. The move from “cost of a workforce” to “cost of an output” is where most of the conceptual work happens, and it is easy to perform without noticing. The curvature that is about to appear in the graph was created entirely by this change of horizontal axis.

Draw the graph



Slide 29

Working through it

What is plotted. Output horizontally, total cost vertically. Dots from the table, curve from the function.

Where it starts. At \$200, not at the origin. The fixed bill is owed before the first component exists, so the curve has a positive intercept. That intercept is the fixed cost, readable directly

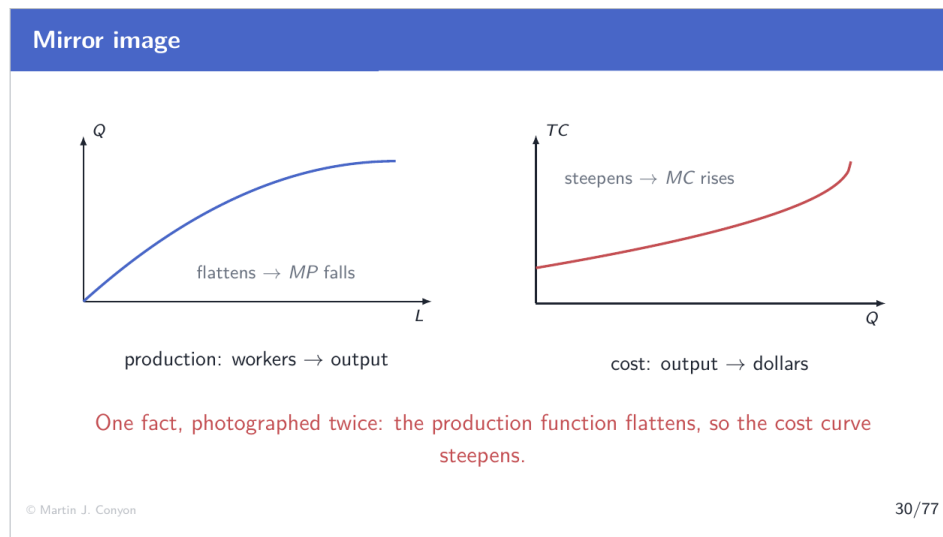
off any total cost curve.

How it rises — and this is the shape to learn. Gently at first, then ever more steeply.

- **At low output** the curve is shallow. Extra components are cheap, because the workers producing them are highly productive.
- **Near capacity** it turns almost vertical. Squeezing out further components takes whole extra hires that each add almost nothing.

What the steepness means. The slope of the total cost curve is the cost of one more unit — marginal cost. So a curve that steepens is marginal cost rising, which is exactly what the next two frames establish numerically.

Mirror image



Slide 30

Working through it

Two pictures, one fact. On the left, output against workers: the production function, *flattening*. On the right, dollars against output: the cost function, *steepening*.

Why they are reflections of each other. Both describe the same relationship between labor and output, read in opposite directions.

- Production asks: given workers, how much output? Its slope is output per worker — marginal product.
- Cost asks: given output, how many workers must be paid? Its slope is (wage $\times$ workers) per unit of output — marginal cost.

Swapping the axes turns one into the other, and a curve that flattens in one orientation steepens in the other.

The single physical fact underneath. Crowding around fixed machines. There is no separate assumption anywhere that costs rise — it is one fact photographed twice.

Why this is worth a frame of its own. Students routinely learn the production function and the cost curves as two separate topics with two separate sets of properties. They are one topic, and every property of the cost curves in Parts 4 and 5 traces back through this mirror.

The cost of one more unit

The cost of one more unit			
One more worker costs \$50 and adds MP units — so each of those units costs			
marginal cost = $\frac{\$50}{MP}$			
Worker	adds (MP)	costs	cost per extra unit
1st	23	\$50	$50/23 = \$2.17$
5th	15	\$50	$50/15 = \$3.33$
7th	11	\$50	$50/11 = \$4.55$
9th	7	\$50	$50/7 = \$7.14$
12th	1	\$50	$50/1 = \$50.00$

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Slide 31

Working through it

The construction. One more worker costs the wage w and produces MP extra units. So each of those units cost

$$MC = \frac{w}{MP}$$

Reading the table.

- 1st worker: adds 23 units for \$50, so $50/23 = \$2.17$ per unit.
- 5th worker: adds 15, so $50/15 = \$3.33$.
- 9th worker: adds 7, so $50/7 = \$7.14$.
- 12th worker: adds 1, so the whole \$50 buys a single component.

The mechanism, in one line. The numerator never moves — the wage is \$50 throughout. The denominator falls steadily, because marginal product diminishes. A fraction with a fixed top and a shrinking bottom can only go one way.

Why marginal cost is not the wage. A common error is to treat the cost of an extra unit as the cost of an extra worker. It is the cost of an extra worker *divided among the units that worker produces*, and that division is where the entire shape of the marginal cost curve comes from.

MC rises because MP falls

MC rises because MP falls

The causal chain, in order:

fixed machines → each worker adds less (*MP falls*) → each unit needs more worker-time → **MC rises**

Marginal cost rises for exactly the reason marginal product falls. Same fact, priced.

This is why short-run cost curves eventually climb — for **every** firm with a fixed input.

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Slide 32

Working through it

The chain, in order.

fixed machines $\Rightarrow MP$ falls $\Rightarrow$ each unit needs more worker-time $\Rightarrow MC$ rises

Reading it link by link. Something is fixed, so extra workers are crowded against it and each adds less. If each worker adds less output, then producing one more unit of output requires more worker-time. And if a unit requires more paid time, it costs more than the unit before it.

What this establishes. Rising marginal cost is *not* a separate assumption smuggled in to make supply curves slope upward. It is diminishing marginal product with a price attached — the same fact, denominated in dollars instead of components.

How general the result is. It holds for *any* firm with a fixed input in the short run, whatever it produces and however it is organized. That generality is what makes the upward-sloping short-run supply curve a prediction rather than a stylized fact.

Where it goes next. Part 4 builds the full family of cost measures around this rising marginal cost, and Part 5 shows what a rising marginal cost does to the averages.

4. Part 4: The Family of Cost Measures

A new firm: the chair workshop

A new firm: the chair workshop

A workshop builds oak chairs. Its daily total cost, in dollars, for Q chairs a day:

$$TC = 100 + 60Q - 12Q^2 + Q^3$$

100 — the daily fixed bill: rent and tool leases, owed at $Q = 0$.

$60Q - 12Q^2 + Q^3$ — the variable bill, shaped so per-unit costs first **fall** (a small crew helps each other) and then **climb** (the bench space runs out).

Same short-run logic — now we need the **per-unit** measures.

Working through it

Why a new firm. The component assembler had a linear cost in workers, which was ideal for showing where rising marginal cost comes from. To display the full family of *per-unit* measures we need a richer cost curve, so the example changes.

The firm. A workshop building oak chairs, with daily total cost in dollars for Q chairs:

$$TC = 100 + 60Q - 12Q^2 + Q^3$$

Reading the pieces.

- **100:** the daily fixed bill — rent and tool leases, owed at $Q = 0$.
- $60Q - 12Q^2 + Q^3$: the variable bill, shaped so that per-unit costs first *fall* and then *climb*.

Why that shape is realistic rather than convenient. At very low output a one-person workshop is inefficient — two people can flip and clamp a frame that one wrestles with alone — so early chairs get cheaper. Past some point the bench space runs out and crowding takes over, exactly as in the assembler. The cubic term is the cheapest algebra that produces both effects at once.

Fixed and variable

Fixed and variable

Fixed costs (FC). Costs that do not vary with output — owed even at $Q = 0$. Here: $FC = 100$.

Variable costs (VC). Costs that vary with output. Here: $VC = 60Q - 12Q^2 + Q^3$.

$$TC = FC + VC$$

Every dollar of total cost is one or the other — **no third category**.

Slide 37

Working through it

The first cut, and it is exhaustive. Every dollar of total cost is either fixed or variable. There is no third category.

- **Fixed costs** (FC) do not vary with output. Here $FC = 100$.
- **Variable costs** (VC) do. Here $VC = 60Q - 12Q^2 + Q^3$.

$$TC = FC + VC$$

The test that settles any classification question. Set $Q = 0$ and ask what remains. In the formula the entire variable expression vanishes at $Q = 0$ and the 100 stays — so 100 is the fixed cost. In words: *what would you still owe if you produced nothing today?*

Applying the test to real items. Rent on a signed lease: fixed. Lumber: variable. A salaried manager who is paid whether or not chairs are built: fixed in the short run. Hourly labor: variable. Note that the same item can be fixed in the short run and variable in the long run — the lease expires eventually — which is exactly the distinction Part 6 turns on.

Why the split matters. The three average measures that follow are built by dividing each piece separately, and they behave very differently.

Plug the numbers in

Plug the numbers in		
Q	$100 + 60Q - 12Q^2 + Q^3$	$TC (\$)$
0	$100 + 0 - 0 + 0$	100
1	$100 + 60 - 12 + 1$	149
2	$100 + 120 - 48 + 8$	180
3	$100 + 180 - 108 + 27$	199
4	$100 + 240 - 192 + 64$	212
5	$100 + 300 - 300 + 125$	225
6	$100 + 360 - 432 + 216$	244
7	$100 + 420 - 588 + 343$	275
8	$100 + 480 - 768 + 512$	324
9	$100 + 540 - 972 + 729$	397
10	$100 + 600 - 1200 + 1000$	500

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Slide 38

Working through it

Every substitution shown, because every later table is built on this column.

$$Q = 3 : 100 + 180 - 108 + 27 = 199$$

$$Q = 7 : 100 + 420 - 588 + 343 = 275 \quad Q = 10 : 100 + 600 - 1200 + 1000 = 500$$

What to notice reading down the total cost column. It rises throughout, but the *steps* between rows are wildly uneven:

- From 4 chairs to 5, total cost rises by only $225 - 212 = \$13$.
- From 9 chairs to 10, it rises by $500 - 397 = \$103$.

Why that unevenness is the whole point. Those step sizes *are* marginal cost. The column already contains the marginal cost curve; the next frames simply read it out and divide it up.

A caution about the third row down. The step from 3 to 4 is \$13 and from 4 to 5 is also \$13. Marginal cost is at its minimum around there — it falls, flattens and then climbs, which is the U-shape that makes the chair workshop a richer example than the assembler.

The per-unit measures: cost of the typical unit

The per-unit measures: cost of the typical unit

Average fixed cost. $AFC = FC/Q$ — the fixed bill spread over the units.

Average variable cost. $AVC = VC/Q$ — the variable bill per unit.

Average total cost. $ATC = TC/Q$ — what the **typical** unit costs.

$ATC = AFC + AVC$ — always, because $TC = FC + VC$.

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Slide 39

Working through it

Three averages, and every one is a division of a column already built.

- **Average fixed cost:** $AFC = FC/Q$ — the fixed bill spread over the units produced.
- **Average variable cost:** $AVC = VC/Q$ — the make-cost of a typical unit.
- **Average total cost:** $ATC = TC/Q$ — what the typical unit costs, all in.

The identity, and why it cannot fail.

$$ATC = AFC + AVC$$

because $TC = FC + VC$, and dividing an identity through by Q preserves it. This is not an empirical finding to be checked against data; it is arithmetic. (It gets checked in a table two frames on anyway, which is the right habit.)

What these measures are for, and what they are not for. They answer “what does a chair cost this workshop on a typical day?” — useful for setting a catalogue price or comparing this year with last.

What they cannot answer. Whether to accept one more order. A typical-unit cost is an average over units already made, and the next unit need not resemble them. That question needs the fourth measure.

Marginal cost: cost of the next unit

Marginal cost: cost of the next unit

Marginal cost. The rise in total cost from one more unit: $MC = \Delta TC / \Delta Q$.

Straight from the TC column:

the 1st chair	149 – 100 = \$49
the 2nd chair	180 – 149 = \$31

ATC : what does the typical unit cost?
 MC : what would **one more** cost?

Different questions — and they can have very different answers.

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40/77

Slide 40

Working through it

The definition. $MC = \Delta TC / \Delta Q$ — the rise in total cost from producing one more unit.

Read straight off the total cost column. No formula, no calculus:

$$\text{1st chair: } 149 - 100 = \$49 \quad \text{2nd chair: } 180 - 149 = \$31$$

Why the first chair costs \$49 and not \$149. The fixed \$100 was owed anyway. Marginal cost measures only what *changed*, so fixed costs never appear in it. This is worth stating explicitly: **marginal cost contains no fixed cost at all**, which is why it is the right number for deciding whether to make one more.

The two questions, deliberately separated.

- ATC : what does the *typical* unit cost?
- MC : what would *one more* cost?

Why they can differ sharply. A workshop might have an average cost of \$40 a chair while the next chair would cost \$60 — and it is the \$60 that should decide whether to accept tomorrow's rush order. Answering an order-desk question with a catalogue number is one of the most expensive mistakes in practical management.

The grand table

The grand table					
Q	TC	$AFC = \frac{100}{Q}$	$AVC = \frac{VC}{Q}$	$ATC = \frac{TC}{Q}$	$MC = \Delta TC$
0	100	—	—	—	—
1	149	100.00	49.00	149.00	49
2	180	50.00	40.00	90.00	31
3	199	33.33	33.00	66.33	19
4	212	25.00	28.00	53.00	13
5	225	20.00	25.00	45.00	13
6	244	16.67	24.00	40.67	19
7	275	14.29	25.00	39.29	31
8	324	12.50	28.00	40.50	49
9	397	11.11	33.00	44.11	73
10	500	10.00	40.00	50.00	103

Audit one row. $Q = 6$: $VC = 244 - 100 = 144$, so $AVC = 144/6 = 24$; $ATC = 244/6 = 40.67$;
 $MC = 244 - 225 = 19$.

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Slide 41

Working through it

What the table is. Five measures, all derived from the single TC column, each by a division or a subtraction.

Auditing row $Q = 6$ from scratch.

- $TC = 244$ (from the cost function).
- $VC = 244 - 100 = 144$, so $AVC = 144/6 = \$24.00$.
- $AFC = 100/6 = \$16.67$.
- $ATC = 244/6 = \$40.67$. Check: $16.67 + 24.00 = 40.67$. ✓
- $MC = 244 - 225 = \$19$ (the sixth chair).

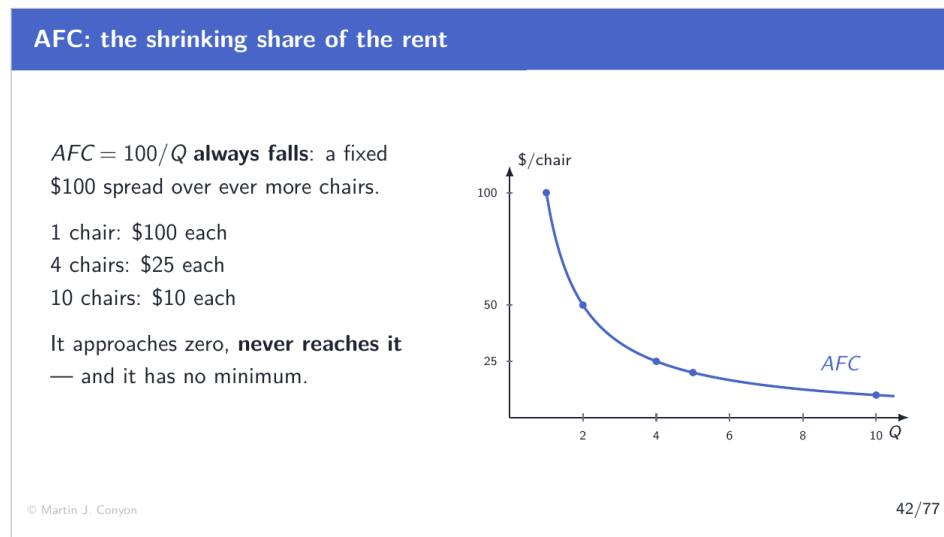
Every other row is the same five operations.

Now scan the columns for shape, because the shapes are the content.

- AFC falls the whole way and never turns.
- AVC falls to \$24 at $Q = 6$, then rises.
- ATC falls to \$39.29 at $Q = 7$, then rises.
- MC falls to \$13 around $Q = 4-5$, then climbs hard to \$103.

The two questions those shapes raise. Why do AVC and ATC turn at *different* outputs, 6 and 7? And why does MC cross each of them exactly at its turn? Part 5 answers both, and the answers are connected.

AFC: the shrinking share of the rent



Slide 42

Working through it

The behavior. $AFC = 100/Q$ falls continuously: \$100 at one chair, \$25 at four, \$10 at ten.

Why it never turns. The numerator is a constant and the denominator only grows, so the quotient can only fall. It approaches zero and never reaches it, and it has no minimum — unlike every other average in the family.

The everyday name for it. “Spreading the overhead”. A business that raises volume without raising rent is doing exactly this, and it is the honest part of the claim that bigger firms have lower costs.

Why this frame is not filler. This quietly falling curve is the answer to the hardest question in Part 5 — why average total cost keeps falling after average variable cost has already turned upward. $ATC = AFC + AVC$, so as long as AFC is falling faster than AVC is rising, their sum still falls. The turn in ATC comes only when the rising side finally overpowers the falling side.

So the thing to carry. AFC has no minimum, and its perpetual decline is a force acting on ATC at every output.

Check the identity

Check the identity

$ATC = AFC + AVC$ — claimed on a definition frame, now **checked in the table**:

Q	AFC	AVC	$AFC + AVC$	ATC
4	25.00	28.00	53.00	53.00 ✓
9	11.11	33.00	44.11	44.11 ✓

Why it must hold: divide $TC = FC + VC$ through by Q .

Habit worth keeping: never trust an identity you have not checked in a row.

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Slide 43

Working through it

The claim, and the check. $ATC = AFC + AVC$ was asserted on a definition frame. Here it is verified against the table.

$$Q = 4 : 25.00 + 28.00 = 53.00 = ATC \checkmark$$

$$Q = 9 : 11.11 + 33.00 = 44.11 = ATC \checkmark$$

Why it must hold. Start from $TC = FC + VC$ and divide every term by Q :

$$\frac{TC}{Q} = \frac{FC}{Q} + \frac{VC}{Q} \quad \text{that is} \quad ATC = AFC + AVC$$

Why check something that must be true. Because the check is not testing the identity — it is testing *your arithmetic*. The identity is guaranteed; your division is not. If the two sides disagree in a row, exactly one of the three numbers in that row is wrong, and you have been told where to look.

The habit being built. Never trust an identity you have not watched hold in at least one row. It costs ten seconds, and it catches errors before they propagate into every subsequent column.

What each column answers

What each column answers		
Measure	The question it answers	Decision it serves
TC	what does the whole day cost?	the budget
ATC	what does the typical chair cost?	catalog pricing
MC	what would one more cost?	take the next order?

Next part: what happens when the **next**-unit number and the **typical**-unit number meet.

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Slide 44

Working through it

Three measures, three distinct jobs.

- TC — what does the whole day cost? The treasurer's number, used for budgeting and financing.
- ATC — what does the typical chair cost? The catalogue number, used for setting a

standing price.

- MC — what would *one more* cost? The order-desk number, used for deciding whether to take the next job.

Why keeping them apart is practical rather than pedantic. Consider a workshop with $ATC = \$40$ and $MC = \$60$ that is offered \$50 a chair for a rush order. Judged against the average, the order looks profitable. Judged against marginal cost — which is the number that actually changes if the order is accepted — it loses \$10 a chair. Only the second comparison is relevant, because only marginal cost measures what accepting the order does to the bill.

What Part 5 is about. Not a new measure, but the *relationship* between the last two. When the cost of the next unit meets the cost of the typical unit, something specific and forced happens — and it is the single most examined result in the whole topic.

5. Part 5: Marginal Pulls Average

Watch the turn

Watch the turn			
ATC falls while MC is below it — and turns where MC passes it:			
Q	MC	ATC	ATC is...
2	31 <	90.00	falling ↓
3	19 <	66.33	falling ↓
4	13 <	53.00	falling ↓
5	13 <	45.00	falling ↓
6	19 <	40.67	falling ↓
7	31 <	39.29	falling ↓ — the bottom
8	49 >	40.50	rising ↑
9	73 >	44.11	rising ↑

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Slide 48

Working through it

Two columns from the grand table, read as a race.

From $Q = 2$ to $Q = 7$. Marginal cost sits *below* average total cost in every row — 31 against 90, 19 against 66, 13 against 53, and so on. And in every one of those rows, *ATC falls*.

At $Q = 7$. The seventh chair costs \$31 against an average of \$39.29, and *ATC* reaches its lowest value: **\$39.29**.

From $Q = 8$ on. The eighth chair costs \$49, now *above* the average — and *ATC* turns and climbs to \$40.50, then \$44.11, and never comes back down.

The pattern to extract. The turn does not happen where marginal cost stops falling. Marginal cost bottoms around $Q = 4-5$ at \$13, and *ATC* carries on falling for two more chairs. The turn happens exactly where the marginal column *overtakes* the average column.

Position, not direction. That distinction — whether *MC* is above or below *ATC*, rather than whether *MC* is rising or falling — is the entire content of the part, and Active Learning 5 is built to catch anyone who misses it.

Your grade average does the same

Your grade average does the same

Three quizzes in, your average is **80** (240 points). The fourth quiz is the *marginal* quiz:

4th quiz	new average	
60	$(240 + 60)/4 = 75$	below — pulls it down
92	$(240 + 92)/4 = 83$	above — pulls it up
80	$(240 + 80)/4 = 80$	equal — it stands still

The average always moves **toward** the newest mark.

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Slide 49

Working through it

Why this analogy and not a diagram. Because you already know the answer, and recognizing that you know it is faster than being taught it.

The setup. Three quizzes in, your average is 80 — that is 240 points over 3 quizzes. The fourth quiz is the *marginal* quiz.

The three cases.

- Score **60** (below the average): $(240 + 60)/4 = 75$. The average falls.
- Score **92** (above the average): $(240 + 92)/4 = 83$. The average rises.
- Score **80** (equal to the average): $(240 + 80)/4 = 80$. The average does not move.

The rule. The average always moves *toward* the newest mark.

What decides the direction. Only whether the new mark is above or below the current average. Note carefully what does *not* matter: whether your scores have been improving or worsening. A student whose scores climb 50, 55, 60, 65 has an average that falls the entire time, because every score is below the running average. That case is exactly the trap in Active Learning 5.

The pulling principle

The pulling principle

Each new unit's cost is a *new mark* entering the average.

- $MC < ATC$: the newcomer is cheap — it **pulls the average down**.
- $MC > ATC$: the newcomer is dear — it **pulls the average up**.
- $MC = ATC$: the average **stands still**.

So MC crosses ATC exactly at the minimum of ATC — a consequence, not an assumption.

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50/77

Slide 50

Working through it

The translation. Each new unit's cost is a new mark entering the average, exactly as each quiz score is.

- $MC < ATC$: the newcomer is cheap — it **pulls the average down**.
- $MC > ATC$: the newcomer is dear — it **pulls the average up**.
- $MC = ATC$: the average **stands still**.

Now follow it to its corner, which is the result. While MC is below ATC , ATC is falling. Once MC is above ATC , ATC is rising. A quantity that falls and then rises has its minimum at the changeover — and the changeover is precisely where $MC = ATC$.

Therefore. MC crosses ATC exactly at the minimum of ATC .

Why that is a consequence rather than an assumption. Nothing was assumed about where the curves cross. Nobody drew them to meet at the bottom. Given only that MC is the cost of the newest unit and ATC is the average of all units, the crossing *has* to be at the minimum. The curves have no choice, and any diagram drawing them otherwise is not merely unconventional — it is wrong.

Check the turn, both directions

Check the turn, both directions

The principle predicts; the table must obey. At the bottom, $ATC = 39.29$ at $Q = 7$:

- **Into the bottom:** the 7th chair cost $31 < 40.67$ (the average it joined) — so it had to pull ATC down: $40.67 \rightarrow 39.29$. ✓
- **Out of the bottom:** the 8th chair costs $49 > 39.29$ — so it had to pull ATC up: $39.29 \rightarrow 40.50$. ✓

Two routes to the same turn — prediction and arithmetic agree. If they had not, one of them would be wrong, and we would go looking.

Working through it

The discipline. A principle predicts; the table must obey. Here the prediction is checked from both sides of the minimum.

Into the bottom. The seventh chair cost \$31. The average it joined was \$40.67 (the ATC at $Q = 6$). Since $31 < 40.67$, the pulling principle says the average must fall. And it does: $40.67 \rightarrow 39.29$. ✓

Out of the bottom. The eighth chair costs \$49. The average it joined was \$39.29. Since $49 > 39.29$, the average must rise. And it does: $39.29 \rightarrow 40.50$. ✓

What has been established, and it is more than a spot-check. The principle was derived by pure reasoning about averages; the table was computed from a cubic cost function. Two entirely independent routes arrive at a turn in the same place. That agreement is what entitles you to believe the result rather than merely recall it.

And the alarm value. Had they disagreed, one of the two would be wrong, and you would know to go looking — which is the same two-route check used throughout this material.

The same pull works on AVC

The same pull works on AVC

MC plays the same game against **average variable cost**:

Q	MC	AVC	AVC is...
5	13 <	25.00	falling ↓
6	19 <	24.00	falling ↓ — the bottom
7	31 >	25.00	rising ↑
8	49 >	28.00	rising ↑

MC crosses AVC at **its** minimum too — same pull, different average.

But wait: AVC bottoms at $Q = 6$, ATC at $Q = 7$. **Why not together?**

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Slide 52

Working through it

Generality first. The pulling principle never asked *which* average was being pulled. It applies to any running average that a new observation joins — so it applies to average variable cost too.

Running the race against AVC .

- $Q = 6$: $MC = 19$ against $AVC = 24$. Below, so AVC falls — and \$24.00 is its minimum.
- $Q = 7$: $MC = 31$ against $AVC = 25$. Above, so AVC rises.

So MC crosses AVC at the minimum of AVC , by exactly the same argument.

The puzzle this creates. AVC bottoms at $Q = 6$. ATC bottoms at $Q = 7$. Both are pulled by the *same* marginal cost column — so why do they turn at different outputs?

Why the question is worth asking rather than waving away. It is not a rounding artefact, and it is not special to these numbers: ATC always bottoms at a *higher* output than AVC , for every firm. The next frame gives the reason, and it is the falling curve from the AFC frame that seemed like filler at the time.

Why does ATC keep falling after AVC has turned?

Why does ATC keep falling after AVC has turned?

Between $Q = 6$ and $Q = 7$, AVC is already rising — yet ATC still falls. Read the pieces off the table:

	ΔAVC	ΔAFC	ΔATC
$Q : 6 \rightarrow 7$	+1.00	-2.38	-1.38 ↓
$Q : 7 \rightarrow 8$	+3.00	-1.79	+1.21 ↑

AFC never stopped falling — it keeps pulling ATC down after AVC has turned. ATC turns only when rising AVC **overpowers** falling AFC .

Two minimums, two outputs — expect to be asked why.

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53/77

Slide 53

Working through it

The apparent contradiction. Between $Q = 6$ and $Q = 7$, AVC is already rising — yet ATC still falls. Since $ATC = AFC + AVC$, one of its two components must be pulling hard in the other direction.

Decomposing the two steps.

- $Q : 6 \rightarrow 7$. $\Delta AVC = +1.00$; $\Delta AFC = -2.38$ (the rent split seven ways instead of six). Net: $\Delta ATC = -1.38$ — still falling.
- $Q : 7 \rightarrow 8$. $\Delta AVC = +3.00$; $\Delta AFC = -1.79$. Net: $\Delta ATC = +1.21$ — now rising.

The mechanism. AFC never stops falling, so it keeps pulling ATC down even after AVC has turned. ATC turns only when the rising AVC finally *overpowers* the falling AFC — which happens one chair later here, and strictly later in general.

Why AVC 's rise eventually wins. AFC 's decline gets weaker with every unit (the gaps shrink: -2.38 , then -1.79), while AVC 's rise gets stronger as crowding accelerates. One force is fading and the other is growing, so the crossover is inevitable.

Expect to be asked. “Why do AVC and ATC have different minimum points?” The answer is one word: AFC .

Efficient scale

Efficient scale

Efficient scale. The quantity at which average total cost is at its minimum.

For the chair workshop: from the table, ATC bottoms at

$$Q = 7 \text{ chairs a day, at } ATC = \$39.29$$

— the output where the workshop makes chairs **as cheaply per chair as it ever can**.
And by the pulling principle, it is where MC meets ATC .

Working through it

The definition. **Efficient scale** is the quantity at which average total cost is at its minimum.

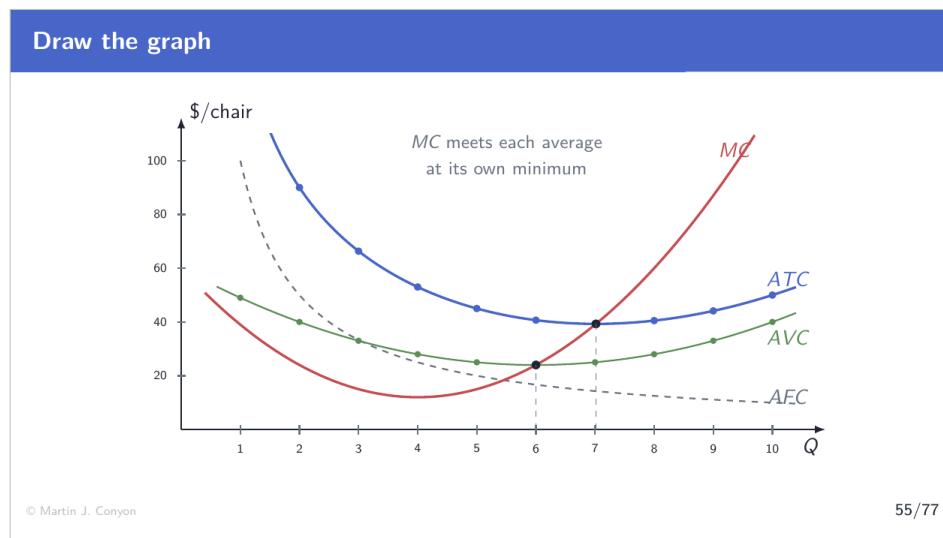
For the chair workshop. $Q = 7$ chairs a day, at $ATC = \$39.29$ — the cheapest per-chair production this shop can achieve on its current premises.

Two ways to locate it, and they must agree. Directly, by finding the smallest number in the ATC column. Or by the pulling principle, by finding where MC crosses ATC . Both give $Q = 7$, as established two frames ago.

What “efficient” does not mean here. Not most profitable, and not the output the firm should choose. Profit depends on price as well as cost, and no price has been mentioned anywhere in this material. A firm facing a high price may rationally produce well beyond efficient scale, accepting a higher average cost in exchange for more units sold.

What it does mean. The point of cheapest averages, given this factory. It becomes important in two later places: it anchors the long-run curve in Part 6, and it is where a competitive industry is driven in the long run — which is why the term is worth fixing precisely now.

Draw the graph



Slide 55

Working through it

What is drawn. All four per-unit measures against output, with the curves generated from the cost function and the dots taken from the table — so the picture cannot contradict the arithmetic.

Reading each curve.

- AFC (dashed, grey): falls forever, approaching the horizontal axis. No minimum.

- *AVC* (green): U-shaped, bottoming at $Q = 6$.
- *ATC* (blue): U-shaped, bottoming at $Q = 7$, and lying above *AVC* everywhere.
- *MC* (red): falls, bottoms early, then climbs steeply.

The gap between ATC and AVC. It is *AFC*, by the identity — so the two curves converge as output grows, because *AFC* is shrinking. They never touch.

The two crossings, which are what an examiner looks for. *MC* passes through the bottom of *AVC* at $Q = 6$, and through the bottom of *ATC* at $Q = 7$. Both are forced by the pulling principle, and a sketch that shows *MC* cutting either average anywhere other than its minimum is incorrect regardless of how neat the drawing is.

Three features every firm shares

Three features every firm shares

Strip away this workshop's numbers; three properties survive:

1. ***MC* eventually rises** — diminishing marginal product prices itself into every next unit.
2. ***ATC* is U-shaped** — falling *AFC* dominates early, rising *AVC* dominates late.
3. ***MC* crosses *ATC* at the minimum of *ATC*** — marginal pulls average; the crossing is forced.

These are structural, not accidents of $100 + 60Q - 12Q^2 + Q^3$.

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56/77

Slide 56

Working through it

The question this frame answers. The chair workshop was one invented firm with one invented cost function. Which parts of the picture are general?

1. *MC* eventually rises. Because in the short run something is fixed, diminishing marginal product operates, and $MC = w/MP$ converts a falling *MP* into a rising *MC*. Any firm with a fixed input, producing anything.

2. ATC is U-shaped. Because $ATC = AFC + AVC$: the falling AFC dominates at low output and the rising AVC dominates at high output. The U-shape is the sum of one force that always falls and one that eventually rises.

3. MC crosses ATC at the minimum of ATC. Because averages are pulled toward the newest mark. This one is pure arithmetic and holds for grade averages and batting averages as readily as for costs.

Why the distinction matters for revision. The particular numbers — \$39.29, seven chairs — are properties of this example and worth nothing on an exam. These three properties are structural, and they are what a question can legitimately test.

Extensions — the pulling principle by calculus

Extensions — the pulling principle by calculus

Everything on this frame is beyond the department's required material. The result was already established from the table; this is the same result, one line of calculus, for those who have met derivatives.

Differentiate $ATC(Q) = TC(Q)/Q$ by the quotient rule:

$$\frac{d(ATC)}{dQ} = \frac{MC \cdot Q - TC}{Q^2} = \frac{MC - ATC}{Q}$$

For $Q > 0$ the sign is the sign of $MC - ATC$: negative when $MC < ATC$, zero when $MC = ATC$, positive when $MC > ATC$ — **the table's pull, in one line.**

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57/77

Slide 57

Working through it

Everything on this frame is beyond the department's required material. The result was already established from the table, twice and in both directions. This is the same result compressed into one line, for readers who have met derivatives.

The differentiation. $ATC(Q) = TC(Q)/Q$, so by the quotient rule

$$\frac{d(ATC)}{dQ} = \frac{MC \cdot Q - TC}{Q^2} = \frac{MC - \frac{TC}{Q}}{Q} = \frac{MC - ATC}{Q}$$

using $TC'(Q) = MC$ and $TC/Q = ATC$.

Reading the sign. For $Q > 0$ the denominator is positive, so the sign of the derivative is the sign of $MC - ATC$:

- $MC < ATC$: derivative negative, ATC falling.
- $MC = ATC$: derivative zero, ATC stationary — the minimum.
- $MC > ATC$: derivative positive, ATC rising.

Which is the pulling principle exactly. The calculus adds no content; it compresses. And it makes plain why the result is unavoidable: the derivative of an average is proportional to the gap between the marginal and the average, so the average can only be stationary where that gap closes.

If you have not met derivatives, nothing is lost. You watched the same result happen chair by chair, and checked it from both sides.

6. Part 6: The Long Run

Fixed today, variable eventually

Fixed today, variable eventually

In the **short run** the factory is given: the only lever is labor, and diminishing *MP* makes costs climb steeply.

In the **long run** everything is a choice: build a bigger plant, automate, redesign the line. **The fixed bill itself becomes variable.**

Short-run cost curves are drawn *for a given factory*. Change the factory, and the whole curve moves.

Working through it

What has been assumed until now. One building, one set of machines. The rent was fixed because a lease was signed and the equipment was bolted down.

Why that cannot last. No lease runs forever. Given enough time the firm can build a bigger plant, buy different machines, or redesign the line entirely — and then the “fixed” bill is simply another choice.

The geometric consequence, which is the real content of the frame. Every short-run cost curve drawn so far belongs to *one particular factory*. Change the factory and the whole curve moves — a different fixed cost, a different *AFC*, a different *AVC*, a different *U*.

So the long-run decision is a different kind of decision. In the short run the firm chooses a *point* on its cost curve. In the long run it chooses *which cost curve to be on*, and then lives on it.

What that means for the next frames. Long-run average cost is not a new kind of cost. It is the answer to: across all the factories the firm could build, what is the cheapest average cost available at each output?

Three factory sizes

Three factory sizes

A firm can build **small**, **medium**, or **large** — each with its own short-run *ATC* curve.

- **Small:** cheap at low output — overwhelmed at high output.
- **Medium:** best in the middle range.
- **Large:** idle capacity at low output — cheapest at high output.

In the **long run** the firm picks the factory; in the **short run** it is stuck with the one it has.

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Slide 62

Working through it

The menu. Small, medium and large plants, each with its own short-run ATC curve.

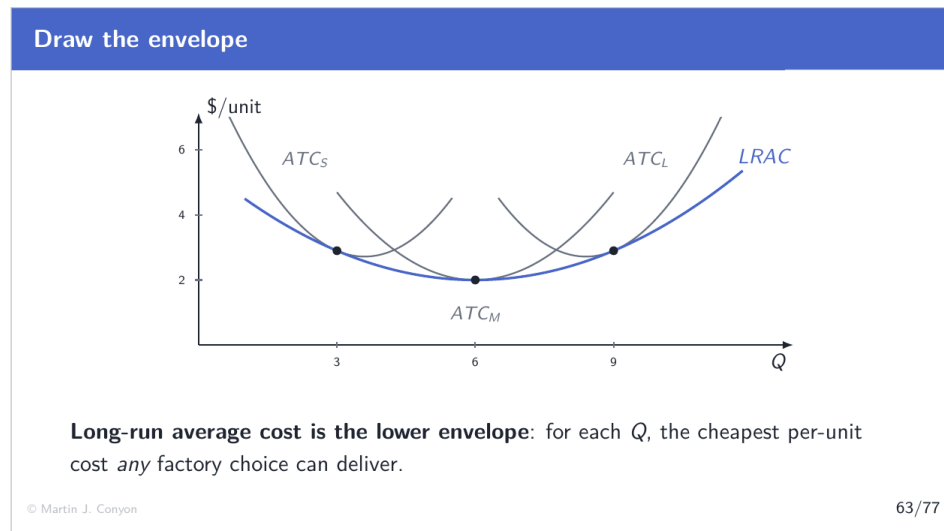
Why each wins somewhere.

- **Small:** little fixed cost, so AFC is low even at modest volume — cheap when orders are thin. But its limited capacity means crowding bites early, so costs explode at high output.
- **Large:** heavy fixed cost, so at low output the rent is spread over too few units and AFC is punishing. At high volume that same rent is spread thin and its capacity is not strained — unbeatable.
- **Medium:** neither extreme, so it rules the middle range.

The trade-off in one line. More capacity means more fixed cost, and fixed cost is only worth paying if there is enough output to spread it over.

The two decisions, kept separate. Which curve to be on is a *long-run* decision, made when the plant is built. Where to sit on it is a *short-run* decision, made daily. A firm that chose well still rides its own curve up and down as orders fluctuate — and if orders change permanently, it faces the long-run decision again.

Draw the envelope



Slide 63

Working through it

What is drawn. Three grey short-run ATC curves, one per factory, and beneath them a blue curve that traces their lower edge.

The question the envelope answers. At each output level: if the firm could choose its plant freshly, what is the cheapest average cost it could achieve? The answer at every output is the lowest of the three grey curves there — and joining those points traces the **long-run average cost curve**.

Why each grey curve touches the envelope exactly once. Each plant is the cheapest option over one range of outputs and beaten elsewhere. Its point of tangency is the volume that plant was built for.

Why every grey curve lies above the envelope everywhere else. Because being stuck with the wrong-sized plant is exactly the situation the long run lets you escape. The vertical distance between a short-run curve and the envelope at any output is the penalty for being in the wrong factory.

With more than three plants. Real firms choose from a near-continuum of sizes, so there are many short-run curves and the envelope becomes smooth — which is how $LRAC$ is normally drawn.

Choosing the size

Choosing the size

Per-unit cost of each factory, read from the three curves:

Q	Small	Medium	Large
2	\$4.00	\$6.80	\$23.20
6	\$5.60	\$2.00	\$5.60
10	\$23.20	\$6.80	\$4.00

Plan for 2 a day: build **small**. For 6: **medium**. For 10: **large**.

In the long run, the firm rides the envelope.

Working through it

The numbers make the choice concrete.

- At $Q = 2$: small **\$4.00**, medium \$6.80, large \$23.20.
- At $Q = 6$: small \$5.60, medium **\$2.00**, large \$5.60.
- At $Q = 10$: small \$23.20, medium \$6.80, large **\$4.00**.

Reading the extremes. At two units a day the large plant charges over \$23 a unit — it is mostly idle, and its rent is being spread over almost nothing. At ten units the small plant charges the same \$23, for the opposite reason: it is being pushed far past its capacity and crowding has taken over. The symmetry of the two numbers is a feature of the example, but the two mechanisms are quite different.

The decision rule. Forecast your output, read across the row, build the cheapest plant.

What “riding the envelope” means. A firm that keeps making this choice correctly — and rebuilding when its output changes permanently — is always on the lowest available curve. That is why the envelope, and not any individual plant’s curve, is the firm’s true long-run cost curve.

Economies and diseconomies of scale

Economies and diseconomies of scale

Economies of scale. *LRAC falls* as output rises.

Constant returns to scale. *LRAC is flat* as output rises.

Diseconomies of scale. *LRAC rises* as output rises.

On our envelope: economies up to $Q = 6$, diseconomies beyond. Many real industries show a **long flat bottom** between the two.

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65/77

Working through it

Vocabulary for the slope of the envelope.

- **Economies of scale:** *LRAC* falls as output rises. Bigger is cheaper per unit.
- **Constant returns to scale:** *LRAC* is flat. Size neither helps nor hurts.
- **Diseconomies of scale:** *LRAC* rises. Bigger has become dearer per unit.

A distinction worth keeping. These describe the *long-run* curve — what happens when the firm rebuilds at a different size. They are not about moving along a short-run curve, where the factory is fixed. Confusing the two is the standard error: a firm whose costs rise because it is straining its existing plant is not experiencing diseconomies of scale.

On our envelope. Economies up to $Q = 6$, diseconomies beyond.

The real-world shape. Many industries show a long *flat bottom* — a wide range of firm sizes that all achieve about the same unit cost. That is why industries frequently contain firms of very different sizes coexisting profitably, which a sharp single-point minimum would not permit.

Why bigger can be cheaper

Why bigger can be cheaper

Specialization. In a large operation, each worker masters one task instead of juggling ten.

Adam Smith opened *The Wealth of Nations* (1776) with a pin factory: ten workers, the pin-making split into distinct operations, producing **tens of thousands of pins a day** — where a lone artisan, doing every step himself, could scarcely make twenty.

Larger scale also justifies **equipment** a small shop could never keep busy.

Working through it

The oldest answer in economics: specialization. In a large operation each worker masters one task instead of juggling ten. Skill rises with repetition, time is not lost switching between tasks, and tools can be dedicated rather than shared.

Smith's pin factory (1776). Adam Smith opened *The Wealth of Nations* with a workshop he had visited: ten workers, the making of a pin split into distinct operations — drawing the wire, straightening, cutting, pointing, grinding, heading — producing tens of thousands of pins a day. A lone artisan performing every step himself could scarcely make twenty.

What the comparison isolates. Nothing about the workers changed. Same people, same tools, same materials. Only the *division of the work* changed, and output per worker rose by orders of magnitude. That is why the example has survived 250 years: it holds everything else constant.

The second source of scale economies. Equipment. A machine that only earns its keep running continuously is unaffordable for a small shop and obvious for a large one. Scale does not make the machine cheaper; it makes there be enough work to keep it busy.

Why bigger eventually costs more

Why bigger eventually costs more

Coordination. As the organization grows:

- management layers multiply,
- information travels slowly and arrives garbled,
- monitoring effort gets harder — and more expensive.

Eventually coordination costs **outgrow** the gains from specialization — and *LRAC* turns upward.

The U-shape of *LRAC* is specialization winning early and coordination losing late.

Working through it

The question. If specialization makes bigger cheaper, why does not every industry collapse into a single enormous firm?

The answer: coordination costs. As an organization grows:

- **Management layers multiply**, so decisions travel further and each layer costs salaries and time.
- **Information travels slowly and arrives garbled**, because it is summarized and re-summarized on the way up.
- **Monitoring gets harder and more expensive**, because it is increasingly difficult to observe who is actually doing what.

Why these grow faster than output. Communication links between people grow roughly with the *square* of the number of people, while output grows roughly in proportion. So coordination is a cost that scales badly by its nature, not through mismanagement.

The result. Past some size, the extra coordination overhead on each additional unit exceeds what further specialization gives back, and *LRAC* turns upward.

The U-shape, explained. Specialization winning early, coordination losing late. Two opposing forces, each dominant over a different range.

Scale shapes the industry

Scale shapes the industry

Compare the **efficient scale** with the **size of the market**:

- Efficient scale **small** relative to demand: many firms fit — a crowded, competitive industry.
- Efficient scale **large** relative to demand: only a few fit — perhaps only **one**.

The shape of *LRAC* is a fundamental determinant of market structure.

Working through it

The comparison that determines market structure. Hold a typical firm's efficient scale — the output at the bottom of $LRAC$ — against the total size of the market.

Case one: efficient scale small relative to demand. A coffee shop, a landscaping firm, a dental practice. Cheapest-per-unit is reached at a volume that is tiny next to total demand, so hundreds or thousands of firms fit. The industry is crowded and competitive, and no firm has any advantage from size.

Case two: efficient scale large relative to demand. A water utility, a rail network, a regional electricity grid. Average cost keeps falling out to volumes that rival the entire market, so one large producer can always undercut any two smaller ones. The market has room for very few firms — sometimes for exactly one.

Why this is a genuine prediction. It says market structure is not an accident of history or regulation but is partly determined by the technology of production, which is observable and measurable independently.

Where it leads. The second case is where natural monopoly comes from, and this curve is the reason monopolies exist in some industries and not others.

7. Part 7: Pulling It Together

The cost concepts at a glance

The cost concepts at a glance		
Term	Definition	Formula
Explicit costs	require a money outlay	—
Implicit costs	forgone alternatives, no outlay	—
Accounting profit	revenue minus explicit costs	$TR - \text{explicit}$
Economic profit	revenue minus all opportunity costs	$TR - (\text{exp} + \text{imp})$
Fixed cost	does not vary with output	FC
Variable cost	varies with output	$VC(Q)$
Total cost	everything	$TC = FC + VC$
Average fixed cost	fixed cost per unit	$AFC = FC/Q$
Average variable cost	variable cost per unit	$AVC = VC/Q$
Average total cost	cost of the typical unit	$ATC = TC/Q$
Marginal cost	cost of the next unit	$MC = \Delta TC / \Delta Q$

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Slide 72

Working through it

What the card contains. Every definition introduced, in one place: the explicit/implicit split, the two profits that follow from it, the fixed/variable split, the three averages, and marginal cost.

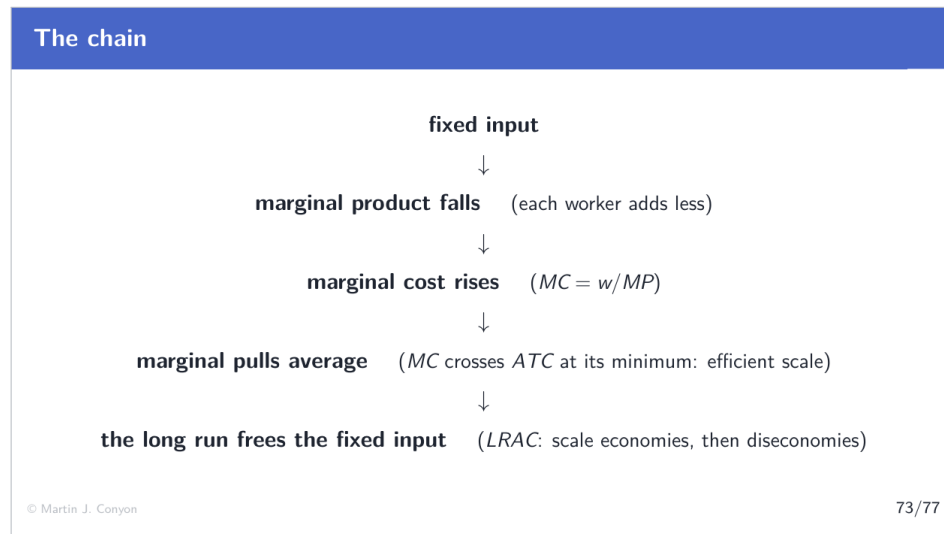
How to revise from it — and the advice is deliberately against memorizing. Do not learn the card. **Rebuild** it. Invent a total cost column of six or seven numbers, choose a fixed cost, and generate every other measure:

- VC by subtracting FC from TC ;
- AFC , AVC , ATC by dividing each by Q ;
- MC by differencing the TC column.

Why rebuilding beats memorizing. Every item below the midline of the card is a division or a subtraction of one column. If you can produce them from a bare cost column in a minute, there is nothing left on the card to forget — and you can handle a question that supplies the columns in an unfamiliar order, which memorization cannot.

The two items above the midline — explicit versus implicit cost, and the two profits — are the only genuine definitions here. Everything else is arithmetic.

The chain



Slide 73

Working through it

Five links, and each one was computed rather than asserted.

Fixed input → **marginal product falls**. Because extra workers are crowded against fixed machines. Established in the component assembler's table: 23, 21, 19, 17.

MP falls → **marginal cost rises**. Because $MC = w/MP$: a constant wage divided by a shrinking number. Established at \$2.17, \$3.33, \$7.14, \$50.00.

MC rises → **marginal pulls average**. Because each unit's cost enters the average like a new quiz mark, so MC crosses each average at that average's minimum. Established in the chair workshop and checked from both directions.

The long run frees the fixed input → **LRAC**. Because the factory becomes a choice, and the firm rides the lower envelope of all the short-run curves — economies of scale, then diseconomies.

Why the chain is worth seeing as a chain. Each link is forced by the one before it. There is exactly one assumption at the top — that something is fixed — and everything else is consequence. That is what makes the cost curves predictions about any firm rather than descriptions of a particular one.

From costs to supply

From costs to supply

The question the cost curves could not yet answer:

How much should the firm produce?

The missing ingredient is the **price** the market pays. The firm will hold price against **marginal cost**, one unit at a time — produce while the unit earns more than it costs.

The cost curves built here are the raw material of the supply curve.

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74/77

Slide 74

Working through it

The question deliberately left unanswered. How much should the firm actually produce? Nothing here can answer it, and the reason is precise: **no price has been mentioned**. Cost curves describe what production costs; they cannot say what output is worth.

The shape of the answer, which you can already anticipate. The firm will hold the market price against *marginal cost*, one unit at a time, and produce while the unit earns more than it costs to make. That is the same marginal comparison used throughout: add while the addition beats its cost.

Why marginal cost and not average cost. Because the decision is about the next unit, and average cost is a fact about units already produced. The order-desk number governs the order-desk decision.

Where this goes. When firms are placed into competitive markets, that comparison traces out the firm's supply curve — and the supply curve turns out to be a segment of the marginal cost curve built here. The curve that has been taken on faith all along is one of the four curves in the last diagram.